

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Murshidha M	Reg. No.:	192571007
Section / Batch:	A	Date:	24/07/2026

Code of Conduct

I Murshidha M _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1**
Binary Search
5 marks
- Q2**
Stack
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Stack - Balancing Symbols
5 marks
- Q6**
Singly Linked List
5 marks
- Q7**
Stack
5 marks
- Q8**
Linked List
5 marks
- Q9**
Linked List
5 marks
- Q10**
Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int binarysearch(int arr[], int n, int key)
3 {
4     int low = 0, high = n - 1;
5     while (low <= high)
6     {
7         int mid = (low + high) / 2;
8         if (arr[mid] == key)
9             return mid;
10        else if (arr[mid] < key)
11            low = mid + 1;
12        else
13            high = mid - 1;
14    }
15    return -1;
16 }
17 int main()
18 {
19     int arr[] = {10, 20, 30, 40, 50, 60};
20     int n = 6;
21     int key = 40;
22     int result = binarysearch(arr, n, key);
23     if (result != -1)
24         printf("Element found at index %d", result);
25     else
26         printf("Element not found");
27     return 0;
28 }
```

[Submitted](#)

Test Input

stdin input (optional)

Output

Element found at index 3

[Pending manual review](#)

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int stack[5];  
3 int top=-1;  
4 int pop() {  
5     if(top== -1) {  
6         printf("stack underflow\n");  
7         return -1;  
8     }  
9     return stack[top--];  
10 }  
11 }  
12 int main() {  
13     int i,n,value;  
14     printf("enter number of elements(max 5): ");  
15     scanf("%d", &n);  
16     if(n>5) {  
17         printf("stack overflow\n");  
18         return 0;  
19     }  
20     printf("enter stack element:\n");  
21     for(i=0; i<n; i++) {  
22         scanf("%d", &stack[i]);  
23     }  
24     top = n-1;  
25     value=pop();  
26     if(value != -1)  
27         printf("popped element:%d\n", value);  
28     return 0;  
29 }  
30 }  
31 }
```

[Run](#)[Save](#)[Submit Fix](#)

Test Input

5

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_19287  
10072A_8a7ec9a2cb8ed4.7025034  
0\main.cc: In function 'main':  
C:\Windows\TEMP\sandbox_19287  
10072A_8a7ec9a2cb8ed4.7025034
```


CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1**
Binary Search
5 marks
- Q2**
Stack
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Stack - Balancing Symbols
5 marks
- Q6**
Singly Linked List
5 marks
- Q7**
Stack
5 marks
- Q8**
Linked List
5 marks
- Q9**
Linked List
5 marks
- Q10**
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[5];  
4 int top;  
5  
6 void init() {  
7     top = -1;  
8 }  
9  
10 int main() {  
11     init();  
12     printf("Top = %d\n", top);  
13     return 0;  
14 }
```

[▶ Run](#)[💾 Save](#)[🚀 Submit Fix](#)

Test Input

stdin input (optional)

Output

Top = -1

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

new.next = head
head = new

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* next;
7 };
8
9 int main() {
10     struct Node* head = NULL;
11     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
12     newNode->data = 10;
13
14     newNode->next = head;
15     head = newNode;
16
17     printf("Node inserted successfully: %d\n", head->data);
18     return 0;
19 }
20
```

▶ Run

💾 Save

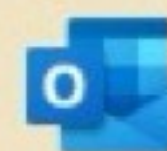
🚀 Submit Fix

Test Input

stdin input (optional)

Output

Node inserted successfully: 10



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node {  
5     int data;  
6     struct Node* next;  
7 };  
8  
9 int main() {  
10     struct Node* head = NULL;  
11     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));  
12     newNode->data = 10;  
13  
14     newNode->next = head;  
15     head = newNode;  
16  
17     printf("Node inserted successfully: %d\n", head->data);  
18     return 0;  
19 }  
20
```

Run

Save

Submit Fix

Test Input

stdin input (optional)



Output

Node inserted successfully: 10

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* next;
7 };
8
9 // Function to reverse the list
10 struct Node* reverseList(struct Node* head) {
11     struct Node* prev = NULL;
12     struct Node* current = head;
13     struct Node* nextNode = NULL;
14
15     while (current != NULL) {
16         nextNode = current->next;
17         current->next = prev;
18         prev = current;
19         current = nextNode;
20     }
21     return prev;
22 }
23
24 // Helper Function to insert a new node at the front
25 void push(struct Node** head_ref, int new_data) {
26     struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
27     new_node->data = new_data;
28     new_node->next = (*head_ref);
29     (*head_ref) = new_node;
30 }
31
32 // Helper Function to print the linked list
33 void printList(struct Node* head) {
34     struct Node* temp = head;
35     while (temp != NULL) {
36         printf("%d -> ", temp->data);
37         temp = temp->next;
38     }
39     printf("NULL\n");
40 }
41
42 int main() {
43     struct Node* head = NULL;
44
45     // Create a list: 4 -> 3 -> 2 -> 1 -> NULL
46     push(&head, 1);
47     push(&head, 2);
48     push(&head, 3);
49     push(&head, 4);
50
51     printf("Original List:\n");
52     printList(head);
53
54     // Reverse the list
55     head = reverseList(head);
56
57     printf("Reversed List:\n");
58     printList(head);
59 }
```

Test Input

stdin input (optional)

Output

```
Original List:
4 -> 3 -> 2 -> 1 -> NULL
Reversed List:
1 -> 2 -> 3 -> 4 -> NULL
```



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 #define MAX 100  
5  
6 int stack[MAX];  
7 int top = -1;  
8 void push(int value) {  
9     if (top == MAX - 1) {  
10         printf("Overflow\n");  
11     } else {  
12         top++;  
13         stack[top] = value;  
14     }  
15 }  
16 int pop() {  
17     if (top == -1) {  
18         printf("Underflow\n");  
19         return -1;  
20     } else {  
21         int data = stack[top];  
22         top--;  
23         return data;  
24     }  
25 }  
26  
27 int main() {  
28     push(10);  
29     push(20);  
30     push(30);  
31  
32     printf("Popped element: %d\n", pop());  
33     printf("Popped element: %d\n", pop());  
34     printf("Popped element: %d\n", pop());  
35  
36     // Testing underflow  
37     pop();  
38  
39     return 0;  
40 }  
41
```

Test Input

stdin input (optional)

Output

```
Popped element: 30  
Popped element: 20  
Popped element: 10  
Underflow
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node* next;
6 };
7
8 int main() {
9     struct Node* head = (struct Node*)malloc(sizeof(struct Node));
10    struct Node* node2 = (struct Node*)malloc(sizeof(struct Node));
11    struct Node* node3 = (struct Node*)malloc(sizeof(struct Node));
12    head->data = 10;
13    head->next = node2;
14
15    node2->data = 20;
16    node2->next = node3;
17
18    node3->data = 30;
19    node3->next = NULL;
20    struct Node* temp = head;
21    while (temp != NULL) {
22        printf("%d\n", temp->data);
23        temp = temp->next;
24    }
25
26    free(head);
27    free(node2);
28    free(node3);
29
30    return 0;
31 }
32
```

Test Input

stdin input (optional)

Output

10
20
30

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == '[' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 char stack[MAX];  
6 int top = -1;  
7  
8 void push(vchar ch) {  
9     stack[++top] = ch;  
10 }  
11 char pop() {  
12     if (top == -1)  
13         return '\0';  
14     return stack[top--];  
15 }  
16  
17 int main() {  
18     char expr[] = "{[()]}";  
19     int i = 0;  
20     while (expr[i] != '\0') {  
21         char ch = expr[i];  
22  
23         if (ch == '[' || ch == '[' || ch == '{') {  
24             push(ch);  
25  
26         else if (ch == ')' || ch == ']' || ch == '}')  
27             pop();  
28     }  
29     if (top == -1)  
30         printf("Balanced\n");  
31     else  
32         printf("Not Balanced\n");  
33     return 0;  
34 }
```

Run

Save

Submit Fix

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\ Sandbox_19257  
10079A_8a7ea03d52bb4.2679495  
0\main.c:8:11: error: unknown  
type name 'vchar'; did you  
mean 'char'?
```


CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 char stack[MAX];  
6 char postfix[MAX];  
7 int top = -1;  
8 int j = 0;  
9  
10 void push(char ch) {  
11     stack[++top] = ch;  
12 }  
13 char pop() {  
14     if (top == -1)  
15         return '\0';  
16     return stack[top--];  
17 }  
18  
19 int main() {  
20     char ch = '(';  
21  
22     push('(');  
23     push('+');  
24     push('*');  
25  
26     if (ch == ')') {  
27         while (top != -1 && stack[top] != '(') {  
28             postfix[j++] = pop();  
29         }  
30         if (top != -1)  
31             pop();  
32     }  
33     postfix[j] = '\0';  
34     printf("postfix: %s\n", postfix);  
35  
36     return 0;  
37 }  
38 }
```

Test Input

stdin input (optional)

Output

postfix:

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1
Stack
5 marks
- Q2
Circular Queue
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Doubly Linked List
5 marks
- Q6
Queue
5 marks
- Q7
Queue - Deque
5 marks
- Q8
Binary Search
5 marks
- Q9
Circular Queue
5 marks
- Q10
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3 #include <ctype.h>  
4  
5 #define MAX 100  
6  
7 int stack[MAX];  
8 int top = -1;  
9  
10 void push(int val) {  
11     if (top < MAX - 1) {  
12         stack[++top] = val;  
13     }  
14 }  
15  
16 int pop() {  
17     if (top >= 0) {  
18         return stack[top--];  
19     }  
20     return 0;  
21 }  
22  
23 int evaluatePostfix(char* expression) {  
24     top = -1; // CRITICAL: Reset top to prevent stack pollution across  
25     int i = 0;  
26  
27     while (expression[i] != '\0') {  
28         if (isdigit(expression[i])) {  
29             push(expression[i] - '0');  
30         } else {  
31             int op1 = pop();  
32             int op2 = pop();  
33             int result = 0;  
34  
35             switch (expression[i]) {  
36                 case '+': result = op2 + op1; break;  
37                 case '-': result = op2 - op1; break;  
38                 case '*': result = op2 * op1; break;  
39                 case '/':  
40                     if (op1 != 0) {  
41                         result = op2 / op1;  
42                     }  
43                     break;  
44             }  
45             push(result);  
46         }  
47         i++;  
48     }  
49     return pop();  
50 }  
51  
52 int main() {
```

Test Input

stdin input (optional)

Output

Security Error: Disallowed code
pattern detected.

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

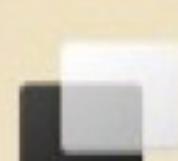
```
1 #include <stdio.h>  
2  
3 #define MAX 5 // Maximum size of the circular queue  
4  
5 int queue[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 // Function to add an element to the queue  
10 void enqueue(int value) {  
11     if ((rear + 1) % MAX == front) {  
12         printf("Queue Overflows\n");  
13         return;  
14     }  
15     if (front == -1) {  
16         front = 0;  
17     }  
18     rear = (rear + 1) % MAX;  
19     queue[rear] = value;  
20 }  
21  
22 // Fixed dequeue function  
23 int dequeue() {  
24     if (front == -1) {  
25         return -1; // Queue Underflow  
26     }  
27     int data = queue[front];  
28  
29     // If it is the last remaining element, reset both pointers to -1  
30     if (front == rear) {  
31         front = -1;  
32         rear = -1;  
33     } else {  
34         front = (front + 1) % MAX;  
35     }  
36  
37     return data;  
38 }  
39  
40 // Function to print the current state of the queue  
41 void display() {  
42     if (front == -1) {  
43         printf("Queue is empty\n");  
44         return;  
45     }  
46     printf("Queue elements: ");  
47     int i = front;  
48     while (1) {  
49         printf("%d ", queue[i]);  
50         if (i == rear) break;  
51         i = (i + 1) % MAX;  
52     }  
53     printf("\n");  
54 }  
55
```

Test Input

stdin input (optional)

Output

```
Queue elements: 10 20 30  
Dequeued: 10  
Dequeued: 20  
Queue elements: 30  
Queue elements: 30 40 50 60
```



CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 // Corrected Recursive Binary Search function  
4 int binarySearch(int arr[], int l, int r, int x) {  
5     // Base Case: If the search space becomes invalid, element is not present  
6     if (l > r) {  
7         return -1;  
8     }  
9  
10    int mid = l + (r - l) / 2;  
11  
12    // If the element is present at the middle itself  
13    if (arr[mid] == x) {  
14        return mid;  
15    }  
16  
17    // If element is smaller than mid, then it can only be in left subarray  
18    if (arr[mid] > x) {  
19        return binarySearch(arr, l, mid - 1, x);  
20    }  
21  
22    // Else the element can only be in right subarray  
23    return binarySearch(arr, mid + 1, r, x);  
24 }  
25  
26 // Main function to test the code  
27 int main() {  
28     int arr[] = {2, 3, 4, 10, 40};  
29     int n = sizeof(arr) / sizeof(arr[0]);  
30     int x = 10;  
31  
32     int result = binarySearch(arr, 0, n - 1, x);  
33  
34     if (result == -1) {  
35         printf("Element is not present in array\n");  
36     } else {  
37         printf("Element is present at index %d\n", result);  
38     }  
39  
40     return 0;  
41 }  
42
```

Test Input

stdin input (optional)

Output

Element is present at index 3

▶ Run

💾 Save

🚩 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1**
Stack
5 marks
- Q2**
Circular Queue
5 marks
- Q3**
Stack
5 marks
- Q4**
Stack
5 marks
- Q5**
Doubly Linked List
5 marks
- Q6**
Queue
5 marks
- Q7**
Queue - Deque
5 marks
- Q8**
Binary Search
5 marks
- Q9**
Circular Queue
5 marks
- Q10**
Stack - Prefix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5 // Maximum size of the Deque  
4  
5 int deque[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 // Function to insert an element at the front  
10 void insertFront(int data) {  
11     // Check if the deque is full  
12     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {  
13         printf("Queue Overflow\n");  
14         return;  
15     }  
16  
17     // If deque is initially empty  
18     if (front == -1) {  
19         front = 0;  
20         rear = 0;  
21     }  
22     // If front is at the first position, wrap around to the end  
23     else if (front == 0) {  
24         front = MAX - 1;  
25     }  
26     // Decrement front normally  
27     else {  
28         front = front - 1;  
29     }  
30  
31     // Assign data to the updated front position  
32     deque[front] = data;  
33     printf("Inserted %d at front. Front is now %d\n", data, front);  
34 }  
35  
36 int main() {  
37     // Test the insertFront function  
38     insertFront(10);  
39     insertFront(20);  
40     insertFront(30);  
41  
42     return 0;  
43 }  
44
```

Test Input

stdin input (optional)

Output

```
Inserted 10 at front. Front  
is now 0  
Inserted 20 at front. Front  
is now 4  
Inserted 30 at front. Front  
is now 3
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 #define MAX 5 // Maximum size of the queue  
5  
6 int queue[MAX];  
7 int front = -1;  
8 int rear = -1;  
9  
10 // Function to insert an element into the queue  
11 void enqueue(int data) {  
12     // 1. Correct Overflow Condition (rear starts at -1, so max index is MAX-1)  
13     if (rear == MAX - 1) {  
14         printf("Queue Overflow\n");  
15         return;  
16     }  
17  
18     // 2. If it's the first element being inserted, initialize front to 0  
19     if (front == -1) {  
20         front = 0;  
21     }  
22  
23     // 3. Pre-increment rear to avoid out-of-bounds access at index -1  
24     queue[++rear] = data;  
25 }  
26  
27 // Function to delete an element from the queue  
28 void dequeue() {  
29     if (front == -1 || front > rear) {  
30         printf("Queue Underflow\n");  
31         return;  
32     }  
33     printf("Dequeued: %d\n", queue[front++]);  
34  
35     // Reset queue pointers if it becomes empty  
36     if (front > rear) {  
37         front = rear = -1;  
38     }  
39 }  
40  
41 // Function to display the queue elements  
42 void display() {  
43     if (front == -1) {  
44         printf("Queue is empty\n");  
45         return;  
46     }  
47     printf("Queue elements: ");  
48     for (int i = front; i <= rear; i++) {  
49         printf("%d ", queue[i]);  
50     }  
51     printf("\n");  
52 }  
53  
54 int main() {  
55     enqueue(10);  
56     enqueue(20);  
57     enqueue(30);  
58     display();  
59     dequeue();  
60     display();  
61 }
```

Test Input

stdin input (optional)

Output

```
Queue elements: 10 20 30  
Dequeued: 10  
Queue elements: 20 30
```


CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

```
POP():  
if top == 0 → Underflow
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 #define MAX 5 // Define the maximum size of the stack  
5  
6 int stack[MAX];  
7 int top = -1; // Global top variable initialised to -1 (empty)  
8  
9 // Corrected POP function  
10 void pop() {  
11     if (top == -1) {  
12         printf("Underflow\n");  
13     } else {  
14         printf("Popped element: %d\n", stack[top]);  
15         top--; // Move top down to remove the element  
16     }  
17 }  
18  
19 // Added main function to fix the compilation error  
20 int main() {  
21     // Example test: attempting to pop from an empty stack  
22     pop();  
23     return 0;  
24 }  
25  
26
```

Run

Save

Submit Fix

Test Input

stdin input (optional)



Output

Underflow

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4 int stack[MAX];
5 int top = -1;
6 void push(int item) {
7     if (top == MAX - 1) {
8         printf("Overflow\n");
9         return;
10    }
11
12    top++;
13    stack[top] = item;
14    printf("Pushed: %d\n", item);
15 }
16
17 int main() {
18     for (int i = 0; i < 6; i++) {
19         push(i + 10);
20     }
21     return 0;
22 }
23
```

Test Input

stdin input (optional)

Output

```
Pushed: 0
Pushed: 10
Pushed: 20
Pushed: 30
Pushed: 40
Overflow
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int main() {
6     int queue[MAX];
7     int front = 0;
8     int rear = 0;
9     int vehicle = 101;
10    if ((rear + 1) % MAX == front) {
11        printf("Queue Full\n");
12    } else {
13        rear = (rear + 1) % MAX;
14        queue[rear] = vehicle;
15        printf("Vehicle %d added to queue at index %d\n", vehicle, rear);
16    }
17
18    return 0;
19 }
20
```

Test Input

stdin input (optional)

Output

```
Vehicle 101 added to queue at index
1
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 extern int top;  
2 int top = -1;  
3 int PEEK(int stack[]) {  
4     if (top < 0) {  
5         return -1;  
6     }  
7     return stack[top];  
8 }  
9 int main() {  
10     int stack[5] = {10, 20, 30, 40, 50};  
11  
12     top = 2;  
13  
14     printf("The top element is: %d\n", PEEK(stack));  
15  
16     return 0;  
17 }  
18  
19  
20  
21
```

Test Input

stdin input (optional)

Output

The top element is: 30

Run

Save

Submit Fix

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

